

Your grade: **100%**

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1. Given the two functions  $f : A \rightarrow B$  and  $g : B \rightarrow C$ , the domain of  $g \circ f$  is  $C$ .

1 / 1 point

- ☐ True
- ☒ False

✔ Nice work

$g \circ f : A \rightarrow C$ , hence, the domain of  $g \circ f$  is equal to  $A$ .

2. Let  $f$  and  $g$  be two functions with  $f(x) = 2x - 1$  and  $g(x) = x^2$ . The function  $(g \circ f)(x)$  is:

1 / 1 point

- ☒  $4x^2 - 4x + 1$
- ☐  $4x^2 + 4x + 1$
- ☐  $2x^2 + 1$
- ☐  $2x^2 - 1$

✔ Nice work

$(g \circ f)(x) = g(2x - 1) = (2x - 1)^2 = 4x^2 - 4x + 1$

3. Given two functions  $f$  and  $g$  with  $f(2) = 3, g(3) = 2, f(3) = 4$  and  $g(2) = 5$ . What is the value of  $(f \circ g)(3)$ ?

1 / 1 point

- ☐ None of these answers
- ☐ 2
- ☒ 3
- ☐ 4

✔ Nice work

$(f \circ g)(3) = f(g(3)) = f(2) = 3$